

# Diffusion Meets Options: Hierarchical Generative Skill Composition for Temporally-Extended Tasks

Zeyu Feng, Hao Luan, Kevin Yuchen Ma, Harold Soh  
{zeyu, haoluan, mayuchen, harold}@comp.nus.edu.sg

## If you're in a hurry

- DOPPLER, a **data-driven** hierarchical reinforcement learning method that plans for linear temporal logic (LTL) instructions;
- A diffusion based approach that learns hierarchical options from offline **non-expert** datasets with diversity sampling;
- Effective **closed-loop** planning by generating and updating plans for long-horizon LTL tasks.



## DOPPLER

Diffusion Option Planning by Progressing LTLs for Effective Receding-horizon control

- Build hierarchical options by learning from offline trajectories with diffusion policy model to provide diverse skills;
- Train by hierarchical deep Q-learning with primitive rewards defined with LTL progression;
- Transfer to unseen LTLs with joint training over state, option and LTL embedding.

- **Product MDP Model with LTL Reward**  
Rewarded when LTL  $\varphi$  is completed.

$$M_{\Psi} = (\mathcal{S}_{\Psi}, \mathcal{A}, p_{\Psi}, r_{\Psi}, \gamma)$$

$$\mathcal{S}_{\Psi} = \mathcal{S} \times \Psi$$

$$r_{\Psi}(\mathbf{s}_t, \varphi, \mathbf{a}_t) = \begin{cases} 1 & \text{if } \text{prog}(\sigma_t, \varphi) = \text{true and } \varphi \neq \text{true} \\ -1 & \text{if } \text{prog}(\sigma_t, \varphi) = \text{false and } \varphi \neq \text{false} \\ 0 & \text{otherwise.} \end{cases}$$

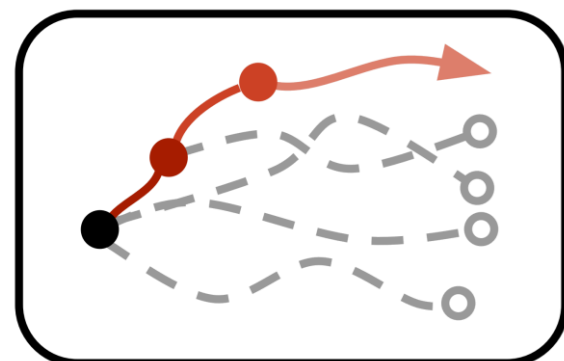
$$p_{\Psi}(\langle \mathbf{s}', \varphi' \rangle | \langle \mathbf{s}, \varphi \rangle, \mathbf{a}) = p(\mathbf{s}' | \mathbf{s}, \mathbf{a}) \text{ if } \varphi' = \text{prog}(L(\mathbf{s}, \mathbf{a}), \varphi) \text{ (and 0 otherwise)}$$

- **Diffusion Option**

Any k-length trajectory in dataset modelled as option.

$$o = \tau_{0:k} = (\mathbf{a}_t, \mathbf{s}_{t+1})_{t=0}^{k-1} \quad \varphi_t = \text{prog}(\sigma_{0:t}, \varphi)$$

$$r_{\Psi}(\mathbf{s}_0, \varphi, o) = \sum_{t=0}^{k-1} \gamma^t r_t \quad r_t = r_{\Psi}(\mathbf{s}_t, \varphi_t, \mathbf{a}_t)$$



Train diffusion and sample option:

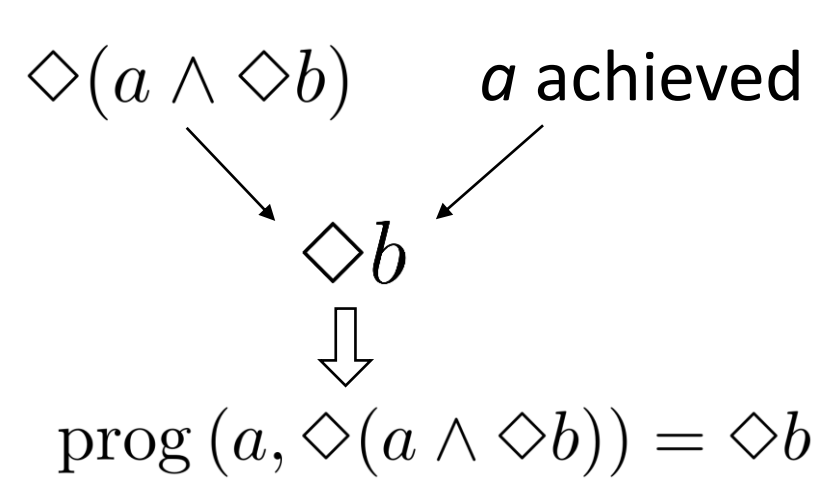
$$\min_{\theta} \mathbb{E}_{i, \tau_{0:k}^i, \tau_{0:k}^0} \left[ \left\| s_{\theta}(\tau_{0:k}^i, i) - \nabla_{\tau_{0:k}^i} \log p(\tau_{0:k}^i | \tau_{0:k}^0) \right\|^2 \right]$$

$$o = \tau_{0:k} \sim \hat{p}_{\theta}(\tau_{0:k} | \mathbf{s})$$

- **Hierarchical Deep Q-Learning**

Minimize:

$$(r_{\Psi}(\mathbf{s}_0, \varphi, o) + \gamma^k \max_{o_k \in \mathcal{O}_{\mathbf{s}_k}} Q(\langle \mathbf{s}_k, \varphi_k \rangle, o_k) - Q(\langle \mathbf{s}_0, \varphi \rangle, o))^2$$



KEY IDEAS

DOPPLER METHOD

DPP SAMPLING

## Diversity Sampling

Generate a batch of M trajectories  $\{\hat{\tau}_{(m)}\}_{m=1}^M$  and each trajectory is conditioned on others.

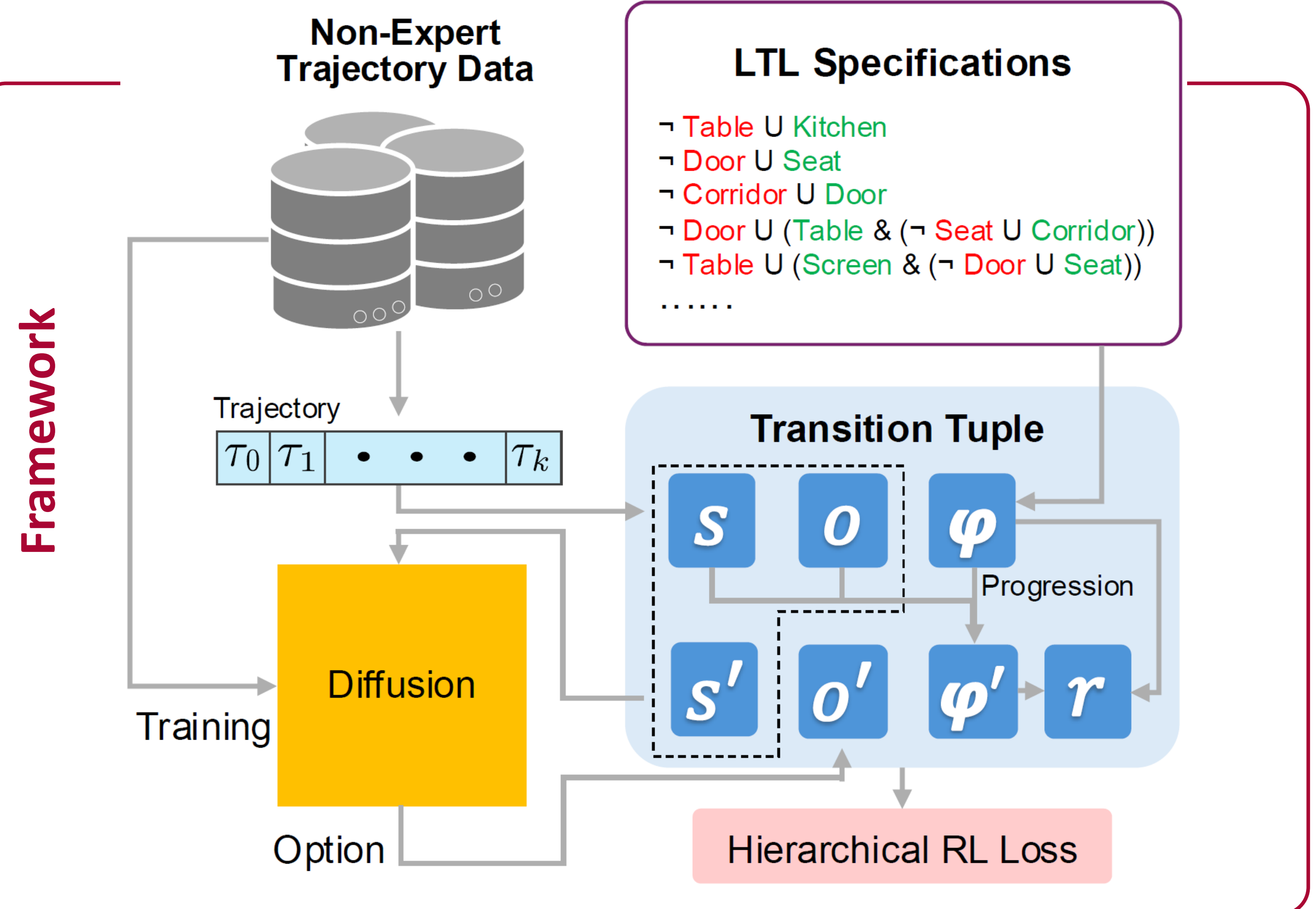
Sample with a conditional score function  $\nabla_{\tau_{(1)}} \log p_i(\tau_{(1)} | \{\hat{\tau}_{(m)}\}_{m=2, \dots, M})$

Similarity in batch measured by Determinantal Point Process (DPP)

$$p(\hat{\tau}_{(1)} | \{\hat{\tau}_{(m)}\}_{m=2}^M) = p(\hat{\tau}_{(1)}) \det(L_M) / Z \quad p(\{\hat{\tau}_{(m)}\}_{m=2}^M | \hat{\tau}_{(1)}) \propto \det(L_M)$$

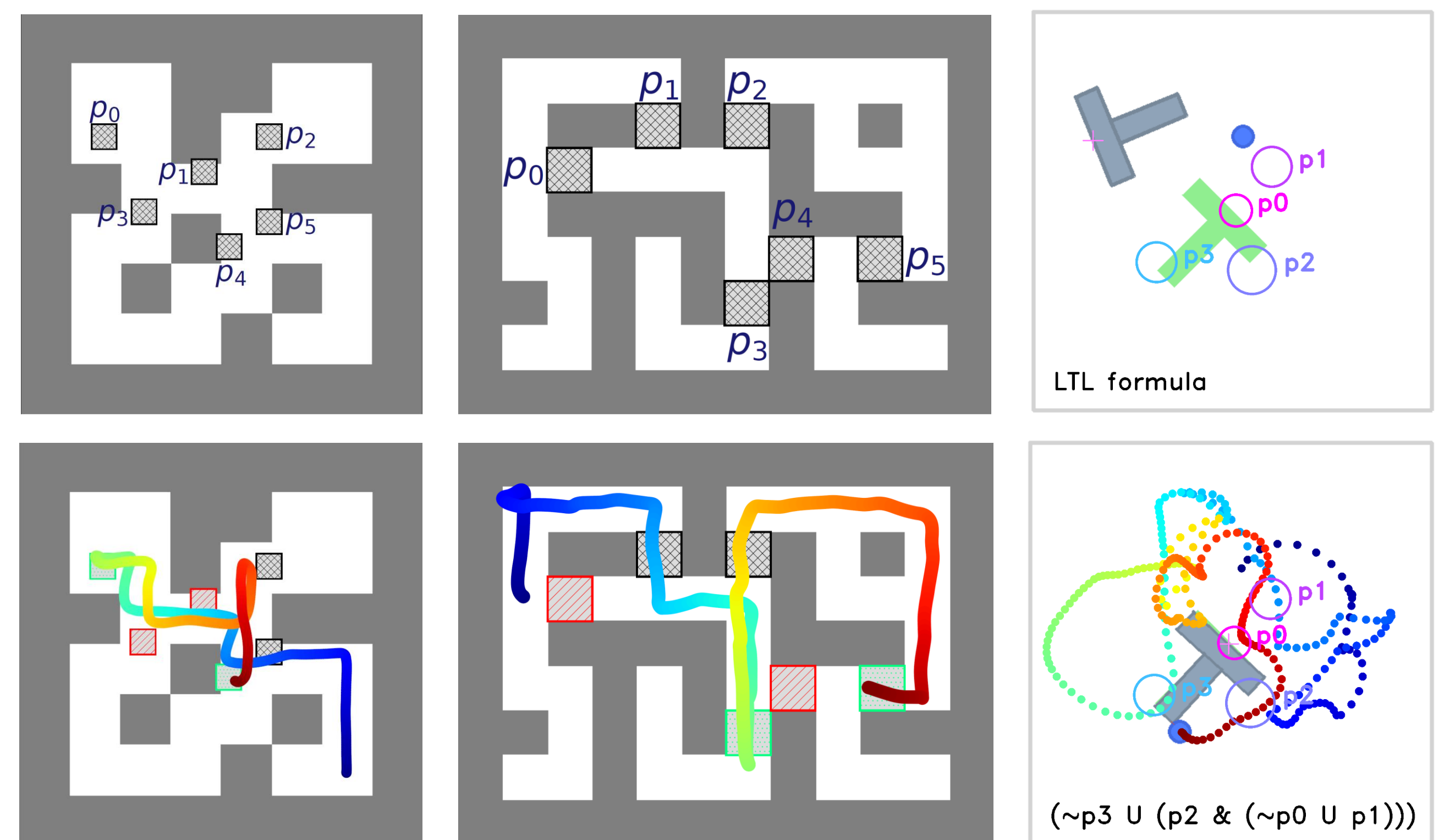
With diffusion posterior sampling:

$$\begin{aligned} \nabla_{\tau_{(1)}} \log p_i(\tau_{(1)} | \{\hat{\tau}_{(m)}\}_{m=2, \dots, M}) &= \nabla_{\tau_{(1)}} \log p_i(\tau_{(1)}) + \nabla_{\tau_{(1)}} \log p(\{\hat{\tau}_{(m)}\}_{m=2, \dots, M} | \tau_{(1)}) \\ &= \nabla_{\tau_{(1)}} \log p_i(\tau_{(1)}) + \nabla_{\tau_{(1)}} \log \int_{\tau_{(1)}} p(\tau_{(1)} | \tau_{(1)}) p(\{\hat{\tau}_{(m)}\}_{m=2, \dots, M} | \tau_{(1)}) d\tau_{(1)} \\ &\approx \nabla_{\tau_{(1)}} \log p_i(\tau_{(1)}) + \nabla_{\tau_{(1)}} \log p(\{\hat{\tau}_{(m)}\}_{m=2, \dots, M} | \hat{\tau}_{(1)}) \\ &= \nabla_{\tau_{(1)}} \log p_i(\tau_{(1)}) + \nabla_{\tau_{(1)}} \log \det(L_M) \end{aligned}$$

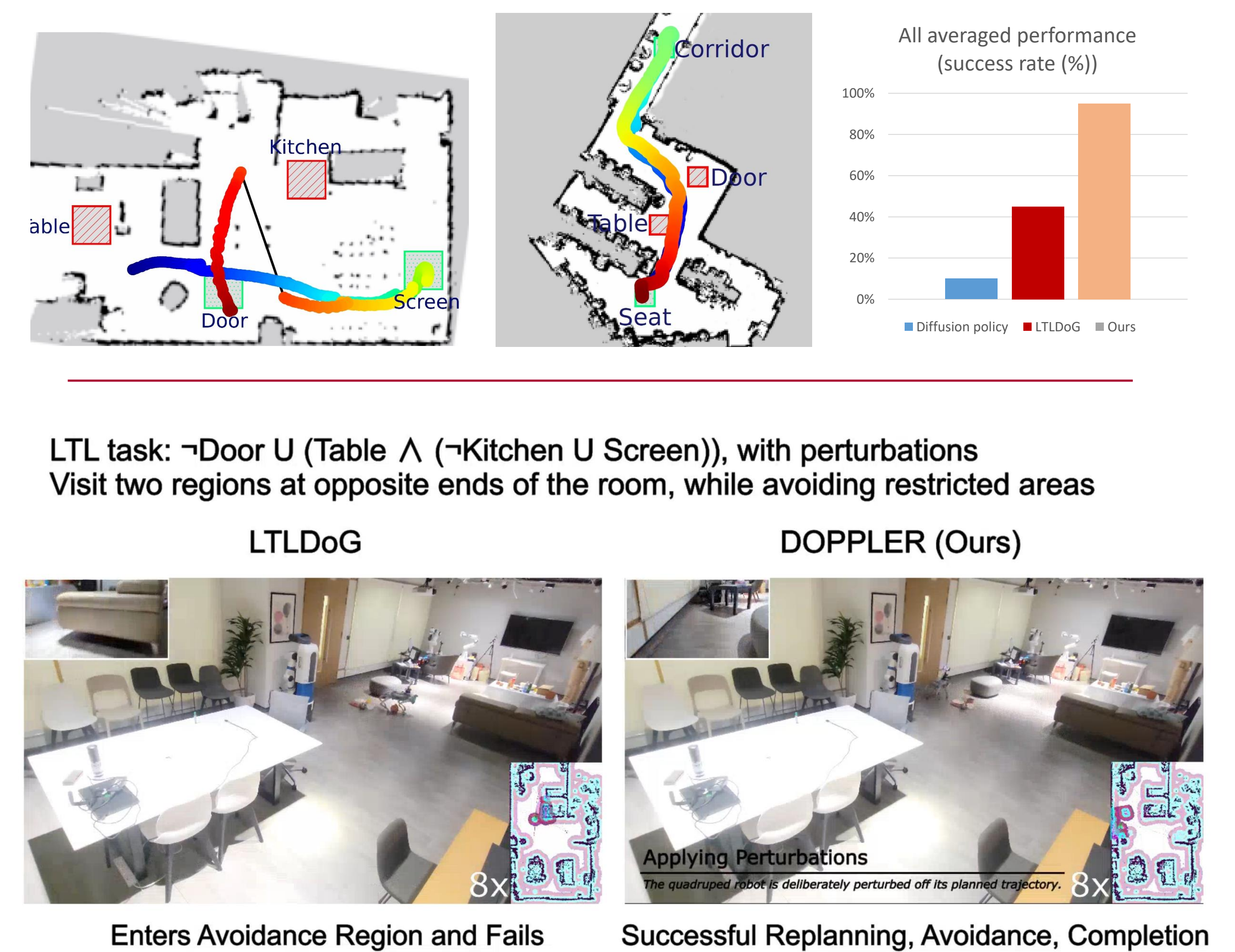


## Simulation & Real Experiments

SIMULATION ENVIRONMENTS



REAL WORLD



## Future Work: What's Next?

- Combine offline learning with limited online adaptation.
- Consider options with different lengths and dynamically select the best horizon to execute.
- Generalize DOPPLER to LTLs where satisfaction theoretically requires infinite sequences to execute.



Check out our web & code  
philiptheother.github.io/doppler/

